



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	FOOD CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. A brief understanding on composition of various foodstuffs 2. Chemical changes in food during processing and storage				
Course Summary	The Course provides an understanding on structure, classification and properties of food nutrients. Comprehend the chemistry of food spoilage, methods of food preservation. Identify the role of natural and artificial food additives and adulterants, their types, and methods of detection. Apply techniques to analyze food samples for adulteration and pesticide residues using gas chromatography, liquid chromatography and mass spectrometry.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand structure, classification, and properties of food nutrients including carbohydrates, proteins, lipids, vitamins and minerals	U	F	Instructor-created exams / Quiz /Assignment
CO2	Comprehend the chemistry of food spoilage, methods of food preservation, and the concept and impact of food packaging and storage	Ap	C	Class test /Assignment /Quiz
CO3	Identify the role of natural and artificial food additives and	An	P	Assignment/ Class test

	adulterants, their types, and methods of detection			
CO4	Apply techniques to analyze food samples for adulteration and pesticide residues using gas chromatography, liquid chromatography and mass spectrometry.	Ap	P	Assignments /Seminar presentation
CO5	Open ended- Evaluate the impact of modern eating habits on health and wellbeing, classifying fast foods, junk foods, instant foods and condiments, and their health effects	E	P	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Food Chemistry- food nutrients		12	20
	1	Carbohydrates: Classification of carbohydrates. Structure and properties of Glucose and Fructose (monosaccharides), Maltose, lactose and sucrose (oligosaccharide) and starch and cellulose (polysaccharide)	3	
	2	Proteins: Introduction to food protein. Structure, classification and physicochemical properties of protein. denaturation, protein determination	3	
	3	Lipids: Classification – Fats and oils – Hydrogenation – Analysis of fats and oils – Acid value, Saponification value and Iodine value.	3	
	4	Minerals: Food minerals, minerals containing Calcium, Iron, Iodine, Sodium and Potassium. Deficiency and toxicity disorders.	2	
	5	Vitamins: Classification, Sources and deficiency diseases.	1	
II	Food Preservation		12	20
	6	Microorganism in food-chemistry of food spoilage: Definition, types of spoilage - physical, enzymatic, chemical and biological spoilage. Mechanism of spoilage.	3	

	7	Methods of food preservation -traditional (drying, smoking, sugaring, freezing, salting, fermentation)	1	
	8	modern methods of food preservation (HPP, PEF, pasteurisation, vacuum packaging, MAP, ohmic heating)	2	
	9	Physical and chemical Additives: – definition, types, Class I and Class II preservatives	2	
	10	Food Packaging and storage - Biodegradable and edible packaging. Environmental concerns, recycling and disposal of packaging waste, Desirable materials for packaging	2	
	11	Shelf life of foods – Definition, intrinsic and extrinsic factors controlling shelf life.	1	
	12	Storage conditions, nutrition value.	1	
III	Food Additives and Adulterants		12	20
	13	Natural and artificial additives for colour and taste- synthetic and natural sweeteners, acidulants, buffering salts, anticaking agents.	4	
	14	Food adulteration - definition and reasons for food adulteration	3	
	15	Methods of adulteration	2	
	16	Common Food Adulterants in Chilli Powder, Tea dust, turmeric powder, milk, vegetable oil, coffee powder	3	
IV	Chemical analysis of food		12	20
	17	Detection of adulteration in various foods-Jam, Tea, Coffee Wheat Flour, Butter, Milk powder, Jelly, Cocoa powder	4	
	18	Analysis of pesticides and insecticides in food	1	
	19	Qualitative Analysis: Gas Chromatography (GC) Liquid Chromatography (LC)	2	
	20	Introduction to HPLC-Separation mechanisms. UV and MS detection	2	
	21	Chromatogram interpretation	2	
	22	Mass Spectrometry (MS)	1	
V	Open Ended- Modern Food Habits		12	20
	23	Definition and health effects of fast foods, instant foods, dehydrated foods, junk foods and condiments - Composition and health effects of chocolates and soft drinks. Harmful effects of modern food habits, Healthy cooking methods	12	

References

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2. John M. Deman, Principles of Food Chemistry, Springer International edition, Third edition, 2007.
3. Meenakshi Paul, Experimental Food Chemistry, Published gene tech books New Delhi,
4. Fenema. R, Food Chemistry, Fourth edition, CRC Press Taylors and Francis groups.
5. B. Srilakshmi, Food Science, 5th Edition, New Age Publishers, New Delhi, 2010.
6. Robertson GL, Food Packaging – Principles and Practice, CRC Press Taylor and Francis Group, 2012
7. Gordon L. Robertson (2012), “Food Packaging: Principles and Practice”, Third Edition, CRC Press.
8. Damodaran, S., Parkin , K L., Fennema, O R.,1996, Fennema’s Food chemistry- 4th edition, CRC press Taylor and Francis Group, New York.
9. Campbell, M K and Farrell, 2006, S O-Biochemistry 5th edition-international student Edition.
10. "Instrumental Methods of Analysis" by Willard, Merritt, Dean, and Settle.
11. "Principles of Instrumental Analysis" by Douglas A. Skoog, F. James Holler, and Stanley R. Crouch.
12. Pearson, D, 2002, the Chemical Analysis of Foods, Churchill Livingstone, New York.
13. O.R. Fennema (2003) Food Chemistry, 3rdEd, Tata McGraw-Hill, New York.
14. Meyer, L H-Food Chemistry. CBS publishers & distributors, New Delhi. 2002
15. Aurand, L.W. and Woods, A.E. 1973. Food Chemistry. AVI, Westport
16. Nielsen, S.S. Introduction to the chemical analysis of foods. Jones and Bartlett Publishers, Boston, London. 2003.
17. Plummer D T (1998) An Introduction to Practical Biochemistry, Third edition, Tata McGraw Hill, New Delhi.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3		2		1		
CO 2	2				1	2	3		2		1		1
CO 3				1	3	1	3		3		1		
CO 4				2	3	1	3		3		1		1
CO 5				3		3	3		2		1		1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓